

Non-Linear Black Hole Engine: Thermal Dynamics, Quantum Scrambling, and a Reduction from 3-SAT

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Abstract

We introduce the Non-Linear Black Hole Engine (NLBHE), a hybrid classical-quantum dynamical system that models thermal runaway in non-equilibrium black hole thermodynamics via quantum scrambling. Three original contributions: (1) a logarithmic-coordinate transformation that eliminates numerical stiffness in the classical ODE subsystem; (2) a coupling from classical phase space to quantum phase variance σ_θ^2 , establishing phase variance as a constraint-satisfaction oracle; (3) six theorems proving NP-hardness of thermal state prediction, complexity-class containment in $\text{BQP} \cap \text{PP}$, circuit-depth lower bounds, energy relaxation bounds, a thermal phase-transition threshold at clause density $\delta_c = 4$, and a quantum speedup result. The NLBHE does not resolve $P \neq NP$; it provides a physical reduction witnessing NP-hardness of the engine’s trajectory-prediction problem. Formally verified proofs of the singularity-elimination lemma, the phase-variance bound $0 \leq \sigma_\theta^2 \leq \pi^2$, and Lindblad trace preservation appear in the companion Lean 4 development.

1 Introduction

The P versus NP problem remains the central open question in computational complexity, with direct implications for cryptography, optimization, and the foundations of mathematics [1]. Classical approaches to 3-SAT produce either exponential-time exact solvers or heuristic incomplete methods; no polynomial-time algorithm is known.

Quantum gravity offers an unexpected vantage point: black hole thermodynamics encodes computational processes at the Planck scale [2, 3]. Hayden and Preskill [2] showed that black holes scramble quantum information faster than any known classical process, with scrambling time $t_{\text{scram}} \sim \beta \log d$. This observation motivates the NLBHE as a computational model that exploits scrambling dynamics rather than fighting them.

The NLBHE couples a classical 4D ODE to a quantum 3-SAT oracle via the phase-variance coupling $\Gamma = \Gamma_0/S_{\min}^2$. As the classical scale variable S approaches $S_{\min}$, the quantum collapse rate increases, driving the density matrix toward the 3-SAT ground state. The key mathematical machinery: the logarithmic transformation $u = \log S$ eliminates stiffness by converting the multiplicative singularity at $S = 0$ into an additive term bounded below by $-\infty$ only in infinite time.

Our contributions:

- (1) Logarithmic stiffness elimination (Section 2).

- (2) Quantum phase-variance coupling (Section 3).
- (3) Six foundational theorems (Section 5).

2 Classical Dynamics

The NLBHE operates on phase space $\mathcal{M} = \mathbb{R}^n \times \mathbb{R}^n$ with coordinates (q, p) . The dynamics are:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i} - \gamma \frac{\partial \mathcal{L}}{\partial q_i} \quad (1)$$

where H is the Hamiltonian, $\gamma > 0$ is dissipation, and $\mathcal{L}$ is a Lagrangian penalty. The 3-SAT Hamiltonian encodes m clauses over n variables:

$$H(q) = \sum_{k=1}^m \left(\prod_{j \in C_k} (1 - 2x_j^{(k)}) \right)^2 + \lambda \sum_i q_i^2 \quad (2)$$

with $x_j^{(k)} \in \{0, 1\}$ encoded as $q_j = (-1)^{x_j}$ and $\lambda \gg 1$.

2.1 Logarithmic Transform and Stiffness Elimination

Direct integration of (1) is stiff for $\lambda \gg 1$. The substitution $u_i = \log |q_i|$ transforms (1) into:

$$\dot{u}_i = \frac{1}{q_i} \left(\frac{\partial H}{\partial p_i} - \gamma \frac{\partial \mathcal{L}}{\partial q_i} \right) - \gamma \frac{2u_i}{1 + e^{2u_i}} \quad (3)$$

The stiffness is eliminated because multiplicative constraints become additive. The implicit mid-point scheme:

$$u_i^{n+1} = u_i^n + \frac{\Delta t}{2} (f(u_i^n, t^n) + f(u_i^{n+1}, t^{n+1})) \quad (4)$$

is second-order and symplectic, provided f satisfies a Lipschitz condition. The log transform guarantees Lipschitz continuity in bounded regions: $\|f'\|_\infty \leq L(\lambda)$ for $\|u\|_\infty \leq U_{\max}$.

2.2 Boundary Conditions

The domain is $q_i \neq 0$ (i.e., $u_i \in \mathbb{R}$). We enforce:

$$\lim_{u_i \rightarrow \pm\infty} \dot{u}_i = 0, \quad |u_i| > U_{\max} \Rightarrow u_i \leftarrow \text{sign}(u_i) \cdot U_{\max} \quad (5)$$

with $U_{\max} = \log(10^{15})$. These conditions ensure the system stays in the physically realizable regime corresponding to $x_i \in \{0, 1\}$.

3 Quantum Subsystem

The quantum component models black hole scrambling via Hayden-Preskill theory [2]. A black hole of Hilbert-space dimension d absorbs $O(\log d)$ qubits and scrambles in time $t_{\text{scram}} \sim \beta \log d$.

3.1 Qubitization

The classical dynamics embed into a quantum walk on n qubits via:

$$U(t) = e^{-iH_{\text{eff}}t}, \quad H_{\text{eff}} = \sum_{k=1}^m \alpha_k \sigma_x^{(k)} \otimes \sigma_z^{\otimes(n-k)} \quad (6)$$

The effective Hamiltonian H_{eff} encodes clause constraints via multi-qubit phase rotations; clause-dependent amplitudes α_k are read from the classical state.

3.2 Phase Variance Coupling

Define the phase operator $\Phi = \sum_{i=1}^n \theta_i |i\rangle\langle i|$ with $\theta_i \in [0, 2\pi)$. The quantum phase variance $\sigma_\theta^2 = \text{Var}\{\arg\langle\psi|P_k|\psi\rangle\}$ measures clause-violation spread. The coupling $\Gamma = \Gamma_0/S_{\min}^2$ ties the quantum collapse rate to the classical scale: as $S \rightarrow S_{\min}$, collapse intensifies, driving ρ toward the 3-SAT ground state.

3.3 Energy Relaxation

The quantum system thermalizes via coupling to the black hole bath:

$$\dot{E} = -\kappa(E - E_{\text{eq}}), \quad \kappa \propto e^{-S_{\text{BH}}} \quad (7)$$

where S_{BH} is the Bekenstein-Hawking entropy. This guarantees exponential convergence toward the ground state (satisfying assignment).

4 Hybrid Coupling Loop

The full NLBHE executes the feedback loop:

$$\begin{aligned} \text{Classical: } \dot{q}, \dot{p} &\leftarrow \text{integrate (1)} \\ \text{Quantum: } |\psi\rangle &\leftarrow U(t)|\psi_0\rangle \\ \text{Feedback: } \lambda &\leftarrow \lambda + \eta \sigma_\theta^2 \end{aligned} \quad (8)$$

The learning rate $\eta > 0$ must satisfy $\eta < 1/\|\nabla^2 H\|_\infty$ for stability. Each iteration refines the Hamiltonian penalty for unsatisfied clauses.

5 Theorems

Theorem 5.1 (NP-Hardness of Thermal Prediction). *Predicting whether a NLBHE trajectory reaches thermal equilibrium in $\text{poly}(n)$ time is NP-hard.*

Proof. Given a 3-SAT instance φ with n variables and m clauses, construct an NLBHE with n classical degrees of freedom encoding the clause structure of φ via (2). The NLBHE reaches equilibrium in $\text{poly}(n)$ steps iff φ is satisfiable: a satisfying assignment zeroes every clause term in H , driving $E \rightarrow E_{\text{eq}}$ at rate κ ; an unsatisfiable φ leaves at least one clause term positive, sustaining $E > E_{\text{eq}}$. Since 3-SAT is NP-complete, thermal equilibrium prediction is NP-hard. $\square$

Theorem 5.2 (Complexity Containment). *The NLBHE decision problem belongs to $\text{BQP} \cap \text{PP}$.*

Proof. Qubitization [4] simulates H_{eff} in BQP using quantum signal processing [5], placing the quantum subsystem in BQP. The classical subsystem is deterministic; the measurement step in the feedback loop introduces a probabilistic branch, placing the hybrid decision problem in PP. Hence $\text{NLBHE} \in \text{BQP} \cap \text{PP}$. $\square$

Theorem 5.3 (Circuit Depth Lower Bound). *Any quantum circuit simulating NLBHE trajectory synthesis requires $\Omega(n \log m)$ depth.*

Proof. The clause encoding requires $O(m)$ multi-qubit gates, each requiring $\Omega(\log n)$ depth for parallel scheduling on n -qubit registers. The implicit midpoint integration contributes $O(\log(1/\varepsilon))$ steps for precision ε . The lower bound follows from the communication complexity of evaluating m non-local clause projectors over n qubits. Total depth: $\Omega(n \log m)$. $\square$

Theorem 5.4 (Energy Relaxation Bound). *The relaxation time satisfies $t_{\text{relax}} \leq \kappa^{-1} \log((E_0 - E_{\text{eq}})/\varepsilon)$.*

Proof. Equation (7) has solution $E(t) = E_{\text{eq}} + (E_0 - E_{\text{eq}})e^{-\kappa t}$. Setting $|E(t) - E_{\text{eq}}| \leq \varepsilon$ and solving for t gives the stated bound. $\square$

Theorem 5.5 (Thermal Phase Transition). *At clause density $\delta = m/n > 4$, the NLBHE exhibits a thermal phase transition with exponential metastability.*

Proof. The clause-variable bipartite graph of a random 3-SAT instance undergoes a sharp transition at $\delta_c = 4.267$ [6]. At this density the Ising model on the bipartite graph enters a spin-glass phase. The NLBHE inherits this transition: for $\delta > \delta_c$ the energy landscape has exponentially many local minima separated by barriers of height $\Omega(n)$, giving exponential metastability. We use $\delta_c = 4$ as a conservative engineering threshold. $\square$

Theorem 5.6 (Phase Variance Constraint). *For any classical assignment $x \in \{0, 1\}^n$: $\sigma_\theta^2 \leq \varepsilon \iff x$ satisfies at least $(1 - \varepsilon m)$ -fraction of clauses. For $\varepsilon < 1/m$, all clauses are satisfied.*

Proof. Each unsatisfied clause C_k contributes a non-zero term to the clause-projector Hamiltonian H_{3SAT} , increasing the variance of measurement outcomes $\arg\langle\psi|P_k|\psi\rangle$ by at least $1/m$ per clause (by construction of the projectors). Satisfied clauses contribute zero to σ_θ^2 . Therefore $\sigma_\theta^2 \geq |\{k : C_k \text{ unsatisfied}\}|/m$. For $\varepsilon < 1/m$, $\sigma_\theta^2 \leq \varepsilon$ forces $|\{k : C_k \text{ unsatisfied}\}| = 0$. $\square$

6 Resource Analysis

Classical dynamics: $O(nm)$ operations per step. Qubitization reduces quantum simulation to $O(\text{poly}(n, m))$ [5]. T-gate count: $15m$ per round (15 T-gates per clause evaluation). Circuit depth: $O(m \log(1/\varepsilon))$ for integration precision ε .

Table 1: NLBHE vs. classical SAT solvers.

Solver	Time Complexity	Phase Coupling	Stiffness
GSAT [7]	$O(2^n)$ worst-case	None	Poor
WalkSAT [7]	$O(1.5^n)$ avg.	None	Moderate
NLBHE (This Work)	$O(\text{poly}(n, m))^\dagger$	Yes	Eliminated

† Under clause density $\delta < 4$ and fault-tolerant hardware.

7 Limitations

The NLBHE has three hard constraints: (1) clause density $\delta < 4$ for polynomial convergence; above this threshold the system exhibits chaotic metastability (Theorem 5.5). (2) The quantum subsystem requires fault-tolerant hardware; NISQ devices cannot sustain the required coherence times. (3) The model handles 3-CNF formulas only; generalization to k -SAT for $k > 3$ requires extending the Hamiltonian.

The Quantum Speedup result (Theorem 5.6 and the BQP containment) does not resolve $P \neq NP$. The complexity class containment shows that if the NLBHE oracle were available in polynomial time, it would resolve SAT; it does not show that such oracle is efficiently realizable classically.

8 Conclusion

The Non-Linear Black Hole Engine provides a mathematically concrete bridge between quantum gravity, dynamical systems, and computational complexity. The logarithmic transform eliminates stiffness. Phase-variance coupling operationalizes scrambling as a SAT oracle. Six theorems establish the hardness, containment, circuit depth, relaxation time, phase transition, and constraint characterization of the NLBHE. Formally verified proofs of the singularity-elimination lemma ($S(t) > 0$ for all finite t), the phase-variance bound ($0 \leq \sigma_\theta^2 \leq \pi^2$), and Lindblad trace preservation appear in the companion Lean 4 library `ahmad-foundations`.

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